

NGUYEN VAN ANH

Backend Engineer Intern

Email: iam.nvanh205@gmail.com | Phone: +84 397 727 903

GitHub: github.com/iamnvanh205 | Portfolio: iamnvanh205.github.io/portfolio/

ABOUT ME

Final-year Software Engineering student focused on backend systems - API design, database architecture, and concurrency control. Built and deployed two backend systems, including a ticketing platform verified correct under 50 concurrent requests with zero oversell. Seeking a backend internship to apply these skills on a real product.

PROJECTS

IT Asset Management System | GitHub: iamnvanh205/itam-system | Demo: itam.vananh.io.vn

Java 21, Spring Boot 3.5, React 19, TypeScript, PostgreSQL, JWT, Docker, JUnit 5, Mockito, Testcontainers

- Built a full IT asset lifecycle system (purchase → assignment → maintenance → audit → disposal) replacing manual Excel tracking, with a multi-branch-ready data model and a 3-step approval workflow (Employee → Manager → IT Staff) across 4 roles.
- Built QR-based asset audits via phone camera with automatic discrepancy detection between scanned and recorded data.
- Secured REST endpoints with JWT authentication, @PreAuthorize-based RBAC, request validation, and CORS configuration; applied Repository/Service layer separation (SOLID) to keep business logic independent from persistence.
- Wrote unit/integration tests (JUnit 5, Mockito, Testcontainers) run on every PR, with full CI/CD: Docker build → push to GHCR → auto-deploy via GitHub Actions. 13 technical docs covering architecture, database design, API, testing, and deployment - github.com/iamnvanh205/itam-system/tree/main/docs

Event Ticketing System | GitHub: iamnvanh205/event-ticketing | Demo: event.vananh.io.vn

Spring Boot, React, TypeScript, PostgreSQL, WebSocket/STOMP, JWT/OAuth2, Docker

- Built an event ticketing platform to solve a specific hard problem: preventing ticket oversell and duplicate check-ins when many users compete for the same limited tickets.
- Designed a reserve-then-confirm booking flow using DB-level pessimistic locking (SELECT FOR UPDATE) to serialize concurrent purchase attempts on the last ticket.
- Validated correctness with concurrency tests: 50 concurrent requests competing for the last ticket, zero oversell, zero duplicate reservation.
- Built a real-time organizer dashboard (WebSocket/STOMP) and QR-based check-in flow, with role-based access across 4 roles (Admin, Organizer, Check-in Staff, Customer).
- Implemented JWT-based authentication via OAuth2, with refresh token handling and logout support through token invalidation.
- Self-hosted the full stack on a VPS using Docker Compose, Nginx reverse proxy, and HTTPS, with the same CI/CD pipeline as ITAM: Docker build → push to GHCR → auto-deploy via GitHub Actions

EDUCATION

Hanoi Open University

Faculty of Information Technology - Major: Software Engineering

2023 - 2027 | GPA: 3.1/4.0

SKILLS

- Languages: Java, JavaScript, TypeScript, SQL
- Frameworks: Spring Boot, Spring Security, Hibernate, JPA
- DevOps: Docker, Git, GitHub Actions, Linux
- Architecture: REST API, Microservices, Clean Architecture, SOLID
- Spoken Languages: Vietnamese (Native), English (proficient in reading technical documentation)